Paper: Facebook's Tectonic Filesystem: Efficiency from Exascale

Jiashu Long

1. What is the primary difference between Blob Storage and Data Warehousing? How does Tectonic address these tenants?
2. Why is disaggregating the metadata necessary for exascale systems? How does Tectonic achieve this? (Distributed Key-Value Store)
3. Data warehousing is in great demand in 2026. Should Tectonic be the standard for designing these systems? What are potential challenges of using Tectonic?

William Hu

1. What are the advantages and trade-offs of using hash-partitioned metadata in Tectonic?
2. How does Tectonic balance performance isolation and resource utilization in a multitenant environment?
3. Why does Tectonic separate metadata into Name, File, and Block layers?

Syed Ikram

1. Why Tectonic’s higher metadata latency an acceptable trade-off for greater scalability compared with HDFS?
2. Growing ML workloads led to Tectonic-Shift, which adds intelligent flash storage to meet higher IOPS demands.
3. Why is Tectonic not suitable for all workloads, despite being designed as a general-purpose filesystem? (Does not support SSD based applications)

Ashray Chaitanya

1. Is Tectonic's decision to use fixed storage-capacity quotas instead of dynamic capacity allocation justified?
2. What is the main strength of Tectonic's TrafficGroup-based approach to multitenancy, and how does it balance isolation with resource utilization ?
3. Do Tectonic's design choices for metadata scalability justify sacrificing certain conventional filesystem features?